

My Java football club management system currently has Player and Salary classes, but it does not yet store match performance details such as goals or assists. Should the Salary class automatically calculate performanceBonus, or should the manager enter a validated bonus manually? Explain the responsibility and data integrity trade-offs.

Thought for a couple of seconds >

For your **current system**, the manager should enter `performanceBonus` manually, with validation.

`Salary` should **not** automatically calculate a performance bonus yet, because the system has no reliable data such as goals, assists, appearances, clean sheets, or match ratings. An automatic calculation without those inputs would be arbitrary and could create incorrect salary records.

Option	Responsibility	Data integrity	Current suitability
Auto-calculate in <code>Salary</code>	<code>Salary</code> needs access to performance rules and match data	Good only when performance data is complete and trusted	Not suitable yet
Manager enters validated bonus	Manager decides bonus based on real-world evaluation/club policy	Depends on validation, but honest about being manually approved	Best choice now

A clean current design:

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```
</> Java
public class Salary {
    private double baseSalary;
    private double performanceBonus;

    public Salary(double baseSalary, double performanceBonus) {
        setBaseSalary(baseSalary);
        setPerformanceBonus(performanceBonus);
    }

    public void setBaseSalary(double baseSalary) {
        if (baseSalary < 0) {
            throw new IllegalArgumentException("Base salary cannot be negative.");
        }
        this.baseSalary = baseSalary;
    }

    public void setPerformanceBonus(double performanceBonus) {
        if (performanceBonus < 0) {
            throw new IllegalArgumentException("Performance bonus cannot be negative.");
        }
        this.performanceBonus = performanceBonus;
    }

    public double calculateTotalSalary() {
        return baseSalary + performanceBonus;
    }
}
```

The manager enters the amount, but the program enforces rules such as:

```
<> Java   
  
if (bonus < 0 || bonus > baseSalary * 0.5) {  
    System.out.println("Bonus must be between 0 and 50% of base salary.");  
}
```

Later, when your project stores match statistics, move the logic outside `Salary`, for example into a `BonusCalculator` or `PayrollService`:

```
<> Java   
  
bonus = goals * GOAL_BONUS + assists * ASSIST_BONUS;
```

That keeps responsibilities clear:

- `Player` stores player information.
- `MatchRecord` or `PerformanceRecord` stores goals, assists, and appearances.
- `BonusCalculator` applies club bonus rules.
- `Salary` stores salary values and calculates the final payable amount.

So the strongest design decision for your audit log/report is: **manual validated bonus now; automatic bonus only after performance data is added.**

```
}  
    performanceBonus = inputNonNegativeDouble(sc, "Performance Bonus: ");  
    calculateTotalSalary();  
}
```

```
public void setPerformanceBonus(double performanceBonus) {  
    if (performanceBonus >= 0) {  
        this.performanceBonus = performanceBonus;  
        calculateTotalSalary();  
    }  
}
```

```
private void calculateTotalSalary() {  
    totalSalary = playerMonthlySalary + performanceBonus;  
}
```